# Voter Cluster Analysis Report Second Draft

by Celine Sachi, Casey Logan

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## Introduction

LLM (Large Language Models) such as Google Gemini, Claude, ChatGPT, are increasingly deployed in domains that requires reasoning about human behavior, social dynamics, and political systems (Wikipedia, 2013). As AI has become part of everyday life in helping people draft emails, summarizing readings, answering questions, etc., these systems are also being increasingly asked to comment on political issues, including elections and voting behavior. Because democratic processes depend on informed public understanding, and the way LLMs reason about and predict voting patterns is not merely a technical issue, yet is a matter of civic importance. Thus, this raises an important question: **How do LLMs reason about voters and political preferences?**

Political prediction and voter modeling are central to campaigns, polling firms, advocacy organizations, and media outlets. In real world, demographic inference models are already used to understand, estimate, and predict elections. Age, race, education level, income, and geographic region are often used to estimate partisan leanings. If LLMs can accurately and responsibly reason about these patterns, then they could serve as a useful educational tools or assist in comparative political analysis. However, if they rely on oversimplified stereotypes or express high confidence in uncertain predictions, then they risk reinforcing misleading narratives. Because elections are foundational to democratic societies, the way AI systems handle politically sensitive information matters.

Recent research from the paper ”On the Dangers of Stochastic Parrots” argue that Large Languange Models can reproduce biases embedded in the training data and generate fluent language without true understanding (Bender, E., McMillan-Major, A., Shmitchell, S., & Gebru, T., 2021). This concern is relevant to voting behavior, such as when the models are asked to predict political preference (Democrat / Republican / Libertarian / many more), (Scot Wilson, 2020) from given voter’s demographic information without recognizing nuance, regional variation, or change over time. Thus, this is prone to stereotyping and overgeneralization.

Beyond academic research, real-world concerns about AI and elections have grown. Policymakers have warned about the role of generative AI in spreading misinformation or shaping public perception during election cycles (Yan, H. Y., Morrow, G., Yang, K.-C., & Wihbey, J., 2025). If a user asks an LLM, “How would this type of voter likely vote?”, the answer may appear authoritative, even if it is speculative. This makes it crucial to understand not only what predictions models make, but how they communicate uncertainty.

This project focuses specifically on how an LLM reasons about voting behavior under different informational conditions. Instead of only testing whether the model leans toward one political party, the study examines how it handles varying levels of detail and ambiguity. The prompts are designed to examine:

- How the model predicts voting preferences based on demographic information.
- Whether it changes its prediction when information is limited.
- Whether it gives consistent answers to the same prompt.
- Whether a single response is accurate and well-calibrated.

| Voter ID | County | Age Group | Gender |
|---|---|---|---|
| 004 | Yakima | 18-24 | Male |
| 005 | Snohomish | 55-64 | Male |
| 006 | King | 55-64 | Female |
| 007 | Pierce | 55-64 | Male |
| 008 | Pierce | 35-44 | Female |
| 009 | Snohomish | 35-44 | Unknown/Other |

Table 1: **sim_wa_voters** sample

Rather than simply asking whether the model is "correct" in a polling sense, this study focuses on the model's accuracy and consistency, as well as determining which variables (voters' information) most influence its predictions.

As LLMs become more integrated into how people access political information, it is important to understand what they predict, how consistently they do so, and what factors drive those predictions. This study aims to provide insight into the reliability and underlying reasoning patterns of LLMs in the context of voting behavior.

# Experimental Design

For this experiment we have two main research questions:

1. How accurate are the predictions made by the LLM?
2. How does the consistency of predictions change when changing both the covariate levels and the covariates provided?

## Research Question 1: Accuracy

To test the accuracy of the predictions, we use data from the Washington Secretary of State office (Reports, Data, and Statistics — WA Secretary of State, 2024). Their government website contains publicly available data related to voter registration, election results, and voter demographics for every year dating back to 2007.

The first step of using this data was to collect demographic information about real Washington voters and use it to create simulated Washington voters. In particular, we used the *Voter Demographics* tables from the *Voter Registration* data. Among this data, we focused on the covariates **County, Age, and Gender**.

To simulate $N = 100$ voters:

1. Sample a county for each voter, using the proportion of voters in each county as probability weights.
2. Use the two-way tables from the *County and Age group* and *County and Gender* data sets to calculate the conditional probabilities $P(\text{age group}|\text{county})$ and $P(\text{gender}|\text{county})$.
3. Sample an age group and gender for each voter, using the above conditional probabilities as probability weights.

The resulting data frame from this simulation will provide us with **sim_wa_voters**, our simulated Washington voters data. A sample of this data can be seen in Table 1.

This data will then be given to each question to generate responses.

The first group of questions relate to party voting habits:

- What party is this voter likely to vote for in the next U.S. presidential election?

- Is this voter likely to vote democrat in the next U.S. presidential election?
- Is this voter likely to vote republican in the next U.S. presidential election?
- Is this voter likely to vote third party in the next U.S. presidential election?

The variation in these questions will provide insight into how the LLM responds when the wording of the question varies. These questions about which party each voter will vote for will provide an understanding of how the LLM interprets general voting behavior based on the given characteristics.

The second group of questions relates to specific items on a ballot. Each question was taken from the Washington 2024 General Election ballot (2024 General Election Voters' Guide — WA Secretary of State, 2024):

1. Federal Candidates
   (a) President/Vice President
   (b) Maria Cantwell or Dr Raul Garcia for U.S. Senator for Washington State
2. Statewide Candidates
   (a) Sal Mungia or Dave Larson for Supreme Court Justice Position #02
   (b) Bob Ferguson or Dave Reichert for Governor
   (c) Mike Pellicciotti or Sharon Hanek for State Treasurer
   (d) Pete Serrano or Nick Brown for Attorney General
3. Measures
   (a) Initiative Measure No. 2066 (concerns regulating energy services)
   (b) Initiative Measure No. 2109 (concerns taxes)
   (c) Initiative Measure No. 2117 (concerns carbon tax credit trading)
   (d) Initiative Measure No. 2124 (concerns state long term care insurance)

These questions will provide insight into how the LLM predicts specific voting behavior, with regard to federal and state voting, based on the given characteristics.

Given the simulated voter data, we then prompted the Google Gemini *gemma-3-27b-it* model to predict how each voter would respond to each question in both groups of questions.

## Research Question 2: Consistency

To test the consistency of the predictions, we simulate voters with a wider range of covariates than we used to test accuracy. In particular the covariates we used were race, age, gender, urbanicity, marital status, number of kids, religion, education, and income.

Levels of each covariate were sampled uniformly to simulate $N = 100$ voters. Voters were also assigned a voter ID.

We then grouped the covariates into **demographics** (race, age, gender, urbanicity), **family** (marital status, number of kids, religion), and **socioeconomic status** (education, income). For each group we created a sub-data frame containing only the covariates for the respective group and the voter ID.

Each data frame will then be given to each question to generate responses.

For consistency testing we add several more questions in addition to those used for accuracy testing.

We add a group of questions taken from a political typology quiz from the Pew Research Center (Pew Research Center, 2021):

- Would this voter rather have small government providing fewer services or bigger government providing more services?
- Does this voter think 'Business corporations make too much profit' or 'Most corporations make a fair and reasonable amount of profit'
- Which comes closer to your view of candidates for political office, even if neither is exactly right? There is at least one candidate who shares most of my views. None of the candidates represent my views well.
- Which of the following statements comes closest to your view? Religion should be kept separate from government policies Government policies should support religious values and beliefs

These questions provide a more comprehensive view of political ideology than those used in the accuracy testing. We are able to use such questions as we are not limited to questions in which we have real data for.

We also ask, "Is the behavior easy to predict?" for each voter. This question will give us a sense of how much the model believes it understands the voters.

Using the same Google Gemini *gemma-3-27b-it* model as before, we provide the voter data each question as a prompt to the model. We then have the model predict a response for the entire data set five times, resulting in 5 predictions per voter, per question.

## Input and Output

To format the prompts, we have to modify the questions to ensure the output is providing the desired results in a format we can work with.

Each question is of the form:

Voter data (voter_id, county, age, gender): *data frame.* For each voter in the voter data, *question* Respond with only *response options.* Can only return unknown sparingly: voter id. prediction

Where *data frame* contains every voter and their covariate information, and *response options* contains the appropriate responses for each question. The output will then return a list containing the prediction for each voter with the respective voter id for identification.

**Example prompts:**

1. Voter data (voter_id, county, age, gender): 1, King, 55-64, Male; 2, Kitsap, 25-34, Male; 3, Thurston, 45-54, Male. For each voter in the voter data, what party is the voter likely to vote for in the next U.S. presidential election? Respond with one word, yes no. Can only return unknown sparingly: voter id. prediction
2. Voter data (voter_id, marital_status, num_kids, religion): 1, widowed, 1, Prefer not to say; 2, widowed, 1, Prefer not to say; 3, married, 4+, Sikh. For each voter in the voter data, would the voter vote for Bob Ferguson or Dave Reichert for Governor in Washington State, in the 2024 general election?, Respond with only the first name of the candidate. Can only return unknown sparingly: voter id. prediction
3. Voter data (voter_id, race, age, gender, urbanicity, marital_status, num_kids, religion, education, income): 1, White, 18-24, unspecified, urban, widowed, 1, Prefer not to say, bachelor, <$25k; 2, Black/African American, 65+, nonbinary, rural, widowed, 1, Prefer not to say, graduate/professional, <$25k. For each voter in the voter data, which of the following statements comes closest to their view? A: Religion should be kept separate from government policies, B: Government policies should support religious values and beliefs. Respond with only A B. Can only return unknown sparingly: voter id. prediction

| | unknown | yes | Total |
|---|---|---|---|
| Democrat | 0 | 49 | 49 |
| Republican | 0 | 49 | 49 |
| Unknown | 2 | 0 | 2 |
| Total | 2 | 98 | 100 |

Table 2: Contingency table for what_party and vote_democrat

| | no | unknown | yes | Total |
|---|---|---|---|---|
| Democrat | 34 | 0 | 15 | 49 |
| Republican | 14 | 0 | 35 | 49 |
| Unknown | 0 | 2 | 0 | 2 |
| Total | 48 | 2 | 50 | 100 |

Table 3: Contingency table for what_party and vote_republican

**Example output for prompt 1**:

1. yes
2. no
3. yes
4. no
5. yes
6. no
7. yes
8. no
9. unknown

# Experiment Analysis

## Accuracy Analysis

### Exploratory Data Analysis

Our preliminary findings indicate a lack of accuracy in voter prediction due to a lack of logical consistency across predictions.

Consider the responses to the prompts (1) "What party is this voter likely to vote for in the next U.S. presidential election", (2) "Is this voter likely to vote democrat in the next U.S. presidential election?", and (3) "Is this voter likely to vote republican in the next U.S. presidential election?"; indicated as what_party, vote_democrat, and vote_republican respectively.

In Tables 2 and 3 we would expect all predictions to either be *Democrat and yes* or *Republican and no* for predictions across prompts to maintain coherence. As we can see in the tables, this is not the case. Table 2 shows we did observe any voters predicted "no" to vote_democrat, but did observe 49 voters predicted to vote Republican but also predicted "yes" to vote_democrat. Similarly in Table 3 we observe 15 voters predicted to

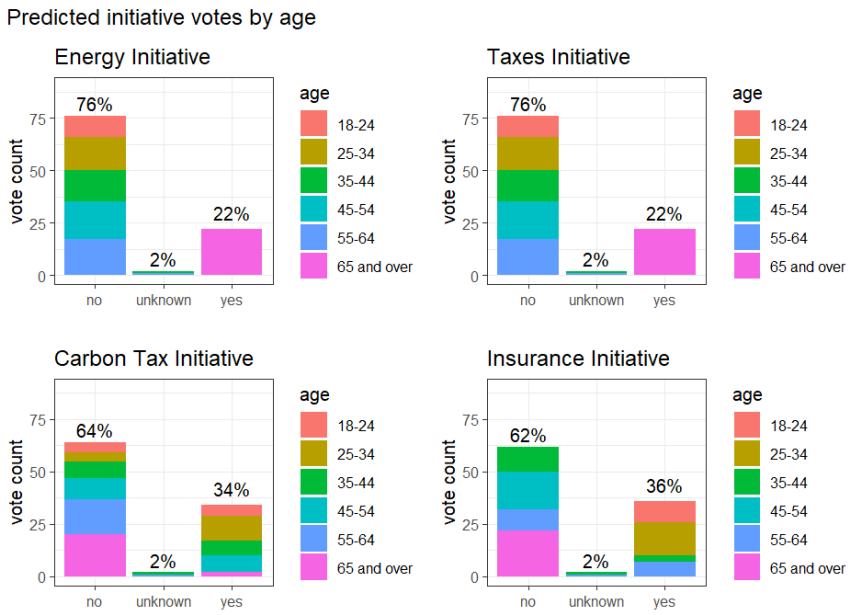



Figure 1: Predicted initiative votes by age

vote Democrat but also predicted "yes" to vote_republican, and 14 voters predicted to vote Republican but also predicted "no" to vote_republican.

In Figure 1 we can see for each initiative, how many voters in each age group were predicted to vote "yes", "no", or "unknown". We know that the energy initiative passed, and the other initiatives did not pass (Washington Secretary of State, 2024). The overwhelming predictions of "no" for each initiative somewhat reflect the true outcomes. However we would expect more predictions of "yes" for the energy initiative, however only 22% of voters were predicted to vote "yes" by the model. Notably, all of the voters predicted "yes" for the energy and taxes initiatives were in the 65 and over age group, but were not highly represented in the same category for the carbon tax and insurance initiatives.

We also see for every initiative, only 2% of voters were predicted as "unknown". Closely inspecting the data we can see that the same 2 voters were predicted "unknown" for every initiative. Specifically, these 2 voters were the only ones simulated to have a gender of "unknown/other". They were also predicted "unknown" for every other prompt provided to the model.

### Analysis Methods

To statistically analyze the accuracy of the data, we will conduct a goodness-of-fit test using Pearson's $\chi^2$.

To calculate the expected value for each combination of covariate levels, we start by calculating the joint probability. We use the chain rule of probability to compute:

$$P(\text{county}, \text{age}, \text{gender}) = P(\text{county}) \cdot P(\text{age}|\text{county}) \cdot P(\text{gender}|\text{county})$$

To calculate the joint probability in this way, we assume that gender and age are conditionally independent given county. We make this key assumption due to the constraints of available data from the Washington Secretary of State office (Reports, Data, and Statistics — WA Secretary of State, 2024).

Then, for each question, we calculate the expected value as:

$$\mathbb{E}[\text{voter prediction}, \text{county}, \text{age}, \text{gender}] = P(\text{county}, \text{age}, \text{gender}) \cdot P(\text{voter prediction}) \cdot N$$

Where $P(\text{voter prediction})$ is the true proportion of votes in the Washington population that voted the sames as the prediction, and $N$ is the population of Washington state. We multiply by $N$ to format the expected values as counts to align with the requirements of the goodness-of-fit test.

For example, consider a simulated voter from Clark county, aged $18 - 24$, who is Female. Suppose the model predicted this voter would vote Bob Ferguson for Washington state governor in the 2024 general election. We calculate the number of voters in Washington state who live in Clark county, are aged $18 - 24$, and are Female is $\approx 17997$ voters. We also know that Bob Ferguson received $40.57\%$ of the votes in the 2024 General Election (Washington Community Alliance, 2021). Therefore,

$$\mathbb{E}[\text{vote Ferguson, Clark County, 18-24, Female}] \approx 7301$$

We then define levels $= \{$all combinations of covariate and prediction levels in data$\}$, $O_i =$ observed count of level combination $i$, and $E_i =$ expected value of level combination $i$.

Given the definitions, the test statistic for our analysis is

$$\chi^2_{obs} = \sum_{i \in \text{levels}} \frac{(O_i - E_i)^2}{E_i}$$

The reference distribution for the test statistic is

$$\chi^2_{obs} \sim \chi^2_{|\text{levels}|-1}$$

Using the simulated voters and predicted voting behavior generated by the model, we can calculate $\chi^2_{obs}$, and use the reference distribution to compute a 2-sided p-value.

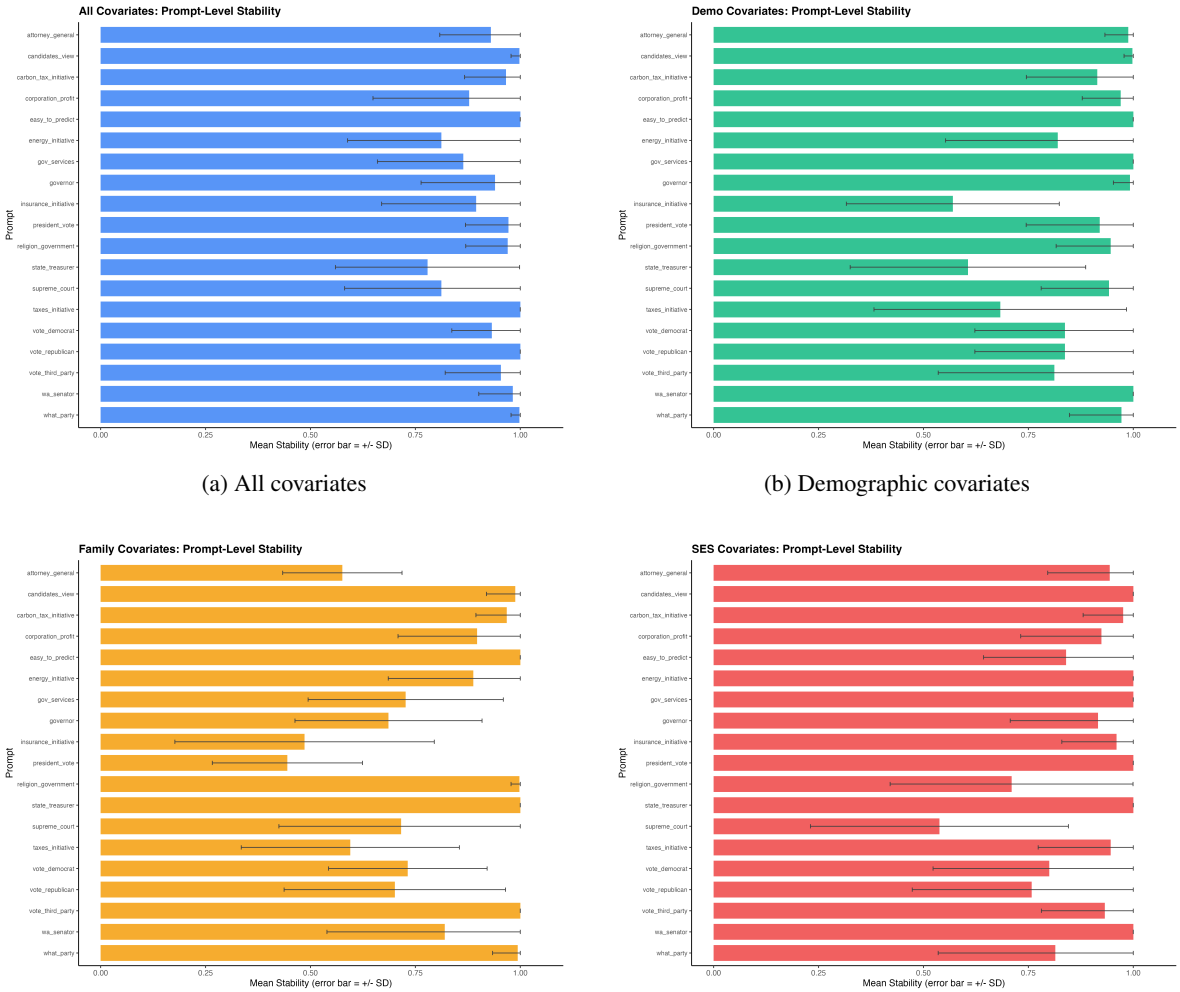
At a significance level of $\alpha = 0.05$, a p-value $< 0.05$ will indicate the generated voter prediction data is significantly different from the expected distribution.

## Consistency Analysis

### Exploratory Data Analysis

In this section, we analyze the consistency of model output across repeated prompt variants using a stability classification method. Initially, cluster analysis was the most intuitive approach (Webster, 2020); however, due to LLM responses being a tiny discrete state space, we do not have many dimensions for response categories and no continuous variables. Therefore, after further research, stability classification is a more suitable analysis approach in this case (Anon, 2023).

Before we begin the stability classification analysis, we explored and examined how stable LLM responses are across pprompts under different covariate settings, also to identify where consistency is strong versus where it fluctuates. Specifically, we explore prompt-level mean stability and variability (standard deviation) within each covariate group to detect early patterns, outliers, and potential prompt effects that may influence later inference.




(a) All covariates

(b) Demographic covariates

(c) Family covariates

(d) SES covariates

Figure 2: Predicted initiative votes by age across model specifications

In the all_covariates plot, prompt-level stability is generally high and tightly clustered, with most prompts showing strong mean stability and relatively small error bars. This indicates that when the model receives the full covariate set, response behavior is not only stable on average but also fairly uniform across prompts. The narrow spread supports the idea that richer contextual information reduces randomness in repeated outputs.

In the demo_covariates plot, stability remains reasonably high overall, but variability across prompts is more visible than in all_covariates. Several prompts show larger SD ranges, suggesting that response consistency under demographic-only information is more sensitive to prompt content. This indicates moderate reliability, but less uniformity than the full-information condition.

In the family_covariates plot, prompt means are lower overall and error bars are often larger, showing both weaker stability and greater fluctuation across prompts. Compared with other groups, this condition appears most heterogeneous, implying that family-level covariates alone provide less consistent guidance for the model's repeated decisions.

In the ses_covariates plot, stability is generally strong and sits between all_covariates and demo/family in terms of variability. Many prompts still produce high mean stability, but there are clearer deviations than in all_covariates. This suggests that SES information contributes useful structure, though not as consistently as the full covariate combination.

Overall, the exploratory patterns provide directional evidence that consistency differs by covariate design,

with all_covariates performing most uniformly and family_covariates showing the most instability. Based on these findings, the next stage of analysis proceeds with the stability classification framework to formally quantify consistency patterns and compare conditions using statistical modeling.

**Analysis Methods**

To statistically analyze the consistency of the data, we began by feeding the LLM five times under the same conditions for each prompt and stored the five responses outputs as one row in a csv file (e.g. ‘attorney_general_0‘ to ‘attorney_general_4‘). We then transformed each row into a numeric stability vector by mapping each response label to a number. For instance, in ‘all_covariates_attorney_general.csv‘, we use ‘pete=1‘ and ‘nick=0‘. Thus, a row like ‘pete,pete,pete,nick,nick‘ becomes $[1, 1, 1, 0, 0]$. This vectorization step creates a standardized representation of response patterns and allows direct comparison across rows and prompts.

Next, we computed a row-level stability score from the five responses. For binary tasks, we count how often each label appears and use the majority share:

$$\text{stability_score}_i = \frac{\max\{n_{\text{pete},i},\, n_{\text{nick},i}\}}{5}.$$

For example, in $[1, 1, 1, 0, 0]$, $n_{\text{pete}} = 3$, $n_{\text{nick}} = 2$, so

$$\text{stability_score} = \frac{3}{5} = 0.60.$$

A perfectly stable row such as ‘pete,pete,pete,pete,pete‘ has score $5/5 = 1.00$, while a split pattern like ‘pete,pete,nick,nick,nick‘ also has $3/5 = 0.60$. We then added this value as a new ‘stability_score‘ column in the CSV.

After scoring each row, we assigned a ‘consistency_level‘ column with three categories:

$$\text{consistency_level} = \begin{cases} \text{high}, & \text{if stability_score} \geq 0.75 \\ \text{medium}, & \text{if } 0.45 \leq \text{stability_score} < 0.75 \\ \text{low}, & \text{if stability_score} < 0.45. \end{cases}$$

Hence, in the attorney_general example, ‘pete,pete,pete,pete,pete‘ is high $(1.00)$, while ‘pete,pete,pete,nick,nick‘ is medium $(0.60)$. Finally, we summarized each prompt file by taking the mean and standard deviation of ‘stability_score‘. Conceptually, higher mean stability indicates that the response is showing a more consistent behavior, while higher standard deviation indicates greater prompt-to-prompt volatility.

At the prompt level, the same trend is visible in concrete examples. In binary-label tasks such as attorney general preference, many rows are perfectly stable (5/5 identical responses), but some rows show partial instability (3/5 versus 2/5). In multi-label tasks (for example, where responses include A, B, and unknown), instability is often driven by fragmentation across labels rather than simple flips between two categories. This indicates that uncertainty-related categories such as unknown may function as a meaningful source of inconsistency rather than random noise.

Furthermore, for deeper analysis, we also compute the overall mean and overall standard deviation across prompts to assess how different covariate information affects the stability of LLM responses. Below is the table of the final summary result:

| Covariate Group | Prompts | Overall Mean | Overall SD |
|---|---|---|---|
| all_covariates | 19 | 0.9306 | 0.0711 |
| demo_covariates | 19 | 0.8846 | 0.1355 |
| family_covariates | 19 | 0.8009 | 0.1878 |
| ses_covariates | 19 | 0.8978 | 0.1255 |

Figure 3: Predicted initiative votes by age

Our final stability summary shows that all_covariates is the most consistent condition: across 19 prompts, it has the highest overall mean stability (0.9306) and the lowest variability (SD = 0.0711), compared with demo_covariates (0.8846, 0.1355), ses_covariates (0.8978, 0.1255), and family_covariates (0.8009, 0.1878) where (mean, standard deviation). This suggests that giving the LLM more complete covariate information leads to more stable and less random responses across repeated prompts.

As a next step, to evaluate whether all_covariates is more consistent than the others, with statistical inference and not just comparing the means (GeeksforGeeks, 2025), we plan to fit an ordinal regression model using consistency_level (low, medium, high) as the outcome and covariate group (all_covariates, demo_covariates, family_covariates, ses_covariates) as the main predictor, with all_covariates as the reference category to directly test whether it yields higher consistency than the other conditions. I will first combine all row-level results into one dataset, verify that the outcome is ordered correctly, and estimate a baseline proportional-odds model; then I will fit alternative specifications (e.g., adding prompt effects or interaction terms) and compare them using AIC and BIC, where lower values indicate a better balance between fit and complexity. After selecting the best model, I will report odds ratios, confidence intervals, and predicted probabilities for each consistency level so the results clearly show how much each covariate configuration changes the likelihood of stable LLM responses.

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